

# Soyo discharge controller

bat2 → grid (RS485)

The Soyo is a grid-tie micro-inverter that discharges the EBox battery (bat2) into the house to keep grid import near zero — the counterpart to the EBox charger. Setpoint  $w \in [0, 900]$  W, cycle 60 s, RS485 frame every 3 s.

## Classification

**Gated P-controller** with night feed-forward and saturation. P-term on grid import (target pcc = 0), no I/D term. Five gates force  $w = 0$ .

## Mathematical model

Gate cascade —  $w = 0$  if any condition holds:

| Gate | Condition                                          |
|------|----------------------------------------------------|
| G1   | stale (inverter data > 3 min old)                  |
| G2   | st ≠ 0 (EBox charging → no simultaneous discharge) |
| G3   | $0 \leq \text{soc2} < 9$ (deep-discharge guard)    |
| G4   | ebox > 200 (bat2 charging)                         |
| G5   | pcc > 200 (PV surplus)                             |

Active law (gate open):

$$w = \begin{cases} \text{sat}_{[0,900]}(k_p(-\text{pcc}) + B_n \mathbf{1}[\text{night}]) & \text{pcc} < -100 \\ B_n \mathbf{1}[\text{night}] + B_d \mathbf{1}[\text{day}] & -100 \leq \text{pcc} \leq 200 \end{cases}$$

The P-term with  $k_p \approx 1$  is a unity feed-forward cancellation: inject as much as is currently imported.  $B_n = 468$  W is the night base load (feed-forward); the dead-band  $[-100, 200]$  W prevents hunting around zero.

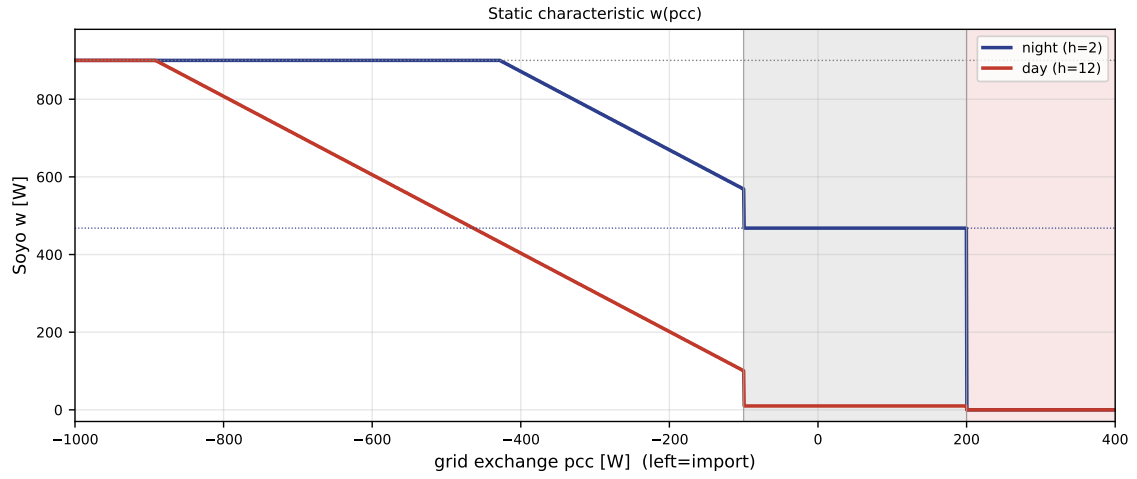
Constants:  $k_p = 1.01$ ,  $B_n = 468$ ,  $B_d = 10$ ,  $w_{\max} = 900$  [W].

TX watchdog (RS485, every 3 s):  $w_{tx} = w \cdot \mathbf{1}[\Delta t \leq 90 \text{ s}]$ . If the setpoint is not refreshed (controller failure), 0 W is sent (fail-safe).

Protocol frame (8 bytes): 24 56 00 21 PH PL 80 CRC with PH=(w»8)&0xFF, PL=w&0xFF, CRC=(264-PH-PL)&0xFF.

## Classification & notes

Pure P-controller on the import error  $e = -\text{pcc}$  (quasi dead-beat per cycle), feed-forward, saturation, dead-band; no integrator → no windup. Charger and discharger tile the pcc axis around 0 and are mutually exclusive (gates st ≠ 0, ebox > 200, pcc > 200). Due to saturation at 900 W the Soyo covers only base load.



Static characteristic: proportional ramp, saturation 900 W, night offset  $B_n$ , dead-band, PV gate.

